Abisko Soil Metagenomics Analysis

Metagenomic Assembly, Binning, and Taxonomic Classification Report

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BINP38 (Applied Work)

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Abstract

This technical report, the complete metagenomic analysis of the soil sample obtained from the Abisko region is described, conducted as an internship project in applied bioinformatics. The high-throughput sequence data were analyzed using the established nf-core/mag pipeline for conducting QC, metagenomic assembly, genome binning, and taxonomy. Using different tools for assembly and binning, this study tried to assemble MAGs as well as analyze the associated microbial community. This technical report describes all techniques used in the process, software engineering and HPC skills gained by me, and taxonomic results observed during the analysis. To ensure reproducibility, all custom scripts, configuration files, and downstream taxonomic reports are present on **Github**.

Contents

Acknowledgements

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1 Introduction

1.1 Metagenomics: Principles and Applications

Metagenomics is a culture-independent molecular tool that involves the direct sequencing of DNA in environmental samples without the requirement of organism culturing. The advent of metagenomics has brought about major changes in microbiology through the ability to study microbial populations in their natural environment. While conventional genomics involves the sequencing of isolated genomes, metagenomics allows for the simultaneous sequencing of thousands of genomes in the sample. In modern metagenomics, there are two major sequencing methods, which are: **short-read sequencing** (typically Illumina technology producing 100–300 bp reads) and **long-read sequencing** (PacBio or Oxford Nanopore producing 10 kbp–100 kbp reads). Short-read sequences offer high accuracy and cost-effectiveness but produce fragmented assemblies; long reads provide superior assembly contiguity but with lower per-base accuracy.

1.2 Permafrost Soils: Study Rationale

Permafrost is permanently frozen ground persisting for two or more consecutive years. Arctic permafrost ecosystems, such as those in Swedish Lapland near Abisko, are of scientific interest due to their role in carbon cycling and potential response to climate change. The soil microbiota in permafrost regions mediates decomposition and carbon transformation processes. Studying microbial community composition across soil layers (surface active layer versus deeper permafrost) provides insights into how microbial metabolism varies with environmental conditions such as temperature, oxygen availability, and organic matter quality.

Internship Objectives

The aims of the current internship project were related to the bioinformatic and computational analyses of soil metagenomic data from the Abisko region. In particular, the aims included:

1. Implementation and optimization of the **nf-core/mag** pipeline for metagenomic assembly and binning on a server by utilizing proper methods of CPU, memory, and disk usage

2. Analyzing soil samples from three study locations, identifying metagenome-assembled genomes (MAGs)

3. Construction of relative species abundance (RSA) plots depicting community profiles from different layers of soils

4. Building a pipeline for analyzing the soil metagenomic data through containerization using the Singularity software

5. Gaining hands-on experience in working with workflow managers, pipeline development, and bioinformatics

2 Methods

2.1 Dataset

2.1.1 Sequencing Data

The sequencing data analyzed comprised 64 soil samples isolated from three different locations within the Abisko area (Kursflaket: 27 samples; Katterjåkk: 10 samples; Storflaket: 27 samples). The sampling was carried out in two different layers at each location (upper and lower layer). Short paired-end reads were obtained through Illumina sequencing (2 × 150 bp), resulting in roughly 1.9 billion reads (equivalent to around 2 TB of compressed data).

2.2 Metagenomics Workflows

All analyses were performed using the `nf-core/mag` pipeline version (v5.0.0) in conjunction with Nextflow (version v21+) for workflow management and Singularity for the deployment of the necessary software packages.

2.2.1 Quality Control and Preprocessing of Reads

The raw reads were subjected to quality control using FastQC v0.11.9+, before and after preprocessing. Trimming and quality filtering were done using fastp v0.23.2+, with any reads less than 50 bases long discarded, while a Phred-quality score of 25 or higher was retained. Normalization of depths was done to minimize extreme coverage differences to avoid bias in assembly and binning, using BBNorm from BBTools at a normalized depth of 100x, with a minimum count of 5.

Critical parameters during preprocessing include:

- Minimum read length: 50 bp
- Quality threshold (Phred): 25
- Normalized depth by BBNorm: 100x (minimum count: 5)

2.2.2 Metagenomic Assembly

Reads were assembled with MEGAHIT (v1.2.9). We used both per-sample assemblies and site/layer co-assemblies (pooling samples within each site and depth layer). Co-assemblies generally improved recovery of metagenome-assembled genomes (MAGs) by increasing effective coverage for low-abundance taxa, and were therefore prioritized for downstream binning.

Assemblies were generated with the `meta-sensitive` preset and a minimum contig length of 1,500 bp. Assembly quality was evaluated with QUAST (v5.2.0), focusing on standard contiguity and composition summaries (e.g., N50/L50, contig length distribution, largest contig, and GC content).

2.2.3 Read Mapping and Contig Coverage Calculation

To support coverage-aware binning, quality-filtered reads were mapped back to the co-assembled contigs with Bowtie2 (v2.4.4, end-to-end mode). Alignments were stored as BAM files and per-contig depth was computed with SAMtools (v1.12). In simplified form:

$$\text{Depth}_{\text{contig}} = \frac{\text{aligned bases to contig}}{\text{contig length}}$$

Depth profiles across samples were summarized into a contig-depth table and used as input to binning tools and abundance summaries.

2.2.4 Genome Binning

Contigs were grouped into MAGs using three complementary binning tools: MetaBAT2 (v2.15), MaxBin2 (v2.2.7), and CONCOCT (v1.1.1). These tools use a combination of sequence composition signals and coverage patterns across samples to cluster contigs that likely originate from the same genome. Each tool was run independently per co-assembly, and the resulting bin sets were combined and refined as described below.

2.2.5 Binning Refinement with DASTool

To reduce redundancy and improve bin quality, DASTool (v1.1.6) was used to integrate the three binning outputs into a consensus set using marker-gene-based scoring and conflict resolution. We used a DASTool quality threshold of 0.3 and required a minimum bin size of 200 kb.

2.2.6 MAG Quality Control and Completeness Assessment

MAG quality was assessed with BUSCO (v5.4.3), which estimates completeness and potential contamination using universal single-copy marker genes (with `auto` lineage selection). Bins with $\geq$10% estimated completeness were retained for taxonomic classification and abundance analysis to avoid discarding partially recovered genomes from low-coverage samples.

For reporting, MAGs were grouped into simple quality tiers:

- High-quality (HQ): $\geq$90% completeness and $\leq$5% contamination
- Medium-quality (MQ): $\geq$50% completeness and $\leq$10% contamination

Additional assembly statistics per bin (e.g., N50/L50 and contig counts) were summarized with QUAST (v5.2.0).

2.2.7 Gene Prediction

Protein-coding genes were predicted on assembled contigs using Prodigal (v2.6.3+). Predicted genes support downstream taxonomy and provide a basis for later functional annotation.

2.2.8 Taxonomic Classification with GTDB-Tk

Taxonomic assignment of MAGs was performed with GTDB-Tk (v2.3.2) against GTDB release 220. GTDB-Tk assigns standardized taxonomy (Domain to Species) by comparing marker-gene profiles to a curated reference and placing MAGs in a reference phylogeny; species-level calls are supported when similarity to a reference is high (e.g., ANI $\geq$ 95%).

To include partial MAGs, classification thresholds were relaxed relative to common defaults (minimum completeness 10%, maximum contamination 50%, minimum aligned amino acids 10%, minimum alignment fraction 0.65), and done with 32 threads.

2.2.9 Abundance Estimation and Relative Sequence Abundance (RSA) Normalization

Following taxonomic classification, bin abundance profiles were calculated across samples using normalized depth information. Relative Sequence Abundance (RSA) was computed per bin i and sample j as:

$$\mathrm{RSA}_{i,j} = \frac{\mathrm{Depth}_{i,j}}{\sum_{i=1}^{n} \mathrm{Depth}_{i,j}} \tag{1}$$

where $\mathrm{Depth}_{i,j}$ is the mean depth of bin i in sample j and the denominator normalizes each sample to a total abundance of 1.0. RSA provides a comparable within-sample composition profile while reducing sensitivity to library size differences.

2.2.10 Publication-Quality Taxonomic Composition Visualization

Taxonomic composition plots were generated with a custom scripts that parses GTDB-Tk summaries, merges taxonomy with abundance estimates, and produces stacked bar plots across standard ranks (Domain through Genus). For clarity, plots were produced both including and excluding unclassified assignments, and were stratified by depth layer when relevant. Output was saved as high-resolution PNG figures suitable for inclusion in the report.

2.3 Computational Infrastructure and Workflow Orchestration

2.3.1 Hardware Configuration

Analyses were executed on a Linux compute node with high core count, large-memory capacity, and SSD-backed local storage. Nextflow work directories were placed on fast local disks to reduce I/O bottlenecks during assembly and binning and the pipeline was executed as a dependency-aware set of processes with explicit resource requests. This enabled parallel execution across samples where possible and supported restart/resume from intermediate checkpoints. All tools were executed inside a Singularity container (`abisko_pipeline.sif`) to ensure consistent software versions and dependencies. The container was configured with bind mounts for inputs, reference databases, and output directories. Finally pipeline resources were controlled via a custom Nextflow configuration file that sets per-process CPU and memory limits and defines database paths. More demanding steps (e.g., co-assembly and taxonomy) were allocated higher resources than lightweight steps.

Quality control summaries from FastQC, QUAST, and BUSCO were aggregated with MultiQC (v1.14). Nextflow also produced standard execution artifacts (timeline, resource usage traces, workflow graph, and logs), enabling traceability of parameters, runtime behavior, and outputs.

2.4 Computational Challenges and Methodological Considerations

2.4.1 Metagenomic Assembly Complexity in Soil Samples

Soil metagenomes are difficult to assemble because communities are highly diverse and abundance is strongly uneven (few dominant taxa and many low-abundance taxa). Repetitive elements and sequencing errors further fragment assemblies. These factors collectively explain the relatively modest contiguity typically observed for soil assemblies compared to isolate genomes.

2.4.2 Bin Quality and Completeness Trade-offs

Binning quality varies across taxa and coverage regimes: some genomes assemble and bin cleanly, while others remain fragmented or show signs of contamination (e.g., duplicated marker genes). Using multiple binners followed by DASTool consensus reduces redundancy and helps select higher-confidence bins, but it can still retain partial MAGs when coverage is limited. The $\geq$10% completeness threshold was chosen to keep partial genomes for broad community-level comparisons.

2.4.3 Taxonomic Novelty and Reference Database Limitations

Some MAGs remained unresolved at fine taxonomic ranks because they are distant from reference genomes, are too incomplete to provide strong phylogenetic signal, or belong to poorly represented soil lineages in current databases. "Unclassified" assignments therefore reflect a mix of genuine novelty and limitations of reference-based classification.

3 Results

The taxonomic analysis was performed on 217 metagenome-assembled bins originating from the Abisko soil metagenomics dataset. Bins were distributed across three sampling locations (SF: 158 bins; KJ: 51 bins; KF: 8 bins) and two depth layers (BOTTOM: 156 bins; TOP: 61 bins). All 217 bins were classified at the phylum level (0.0% unclassified). The following subsections present the layer-wise, cross-layer, location-wise, and cross-location distributions for each taxonomic rank (Phylum, Class, Order, Family, Genus, Species). Relative abundances are computed as the percentage of bins per group; absolute bin counts are shown in parentheses where relevant.

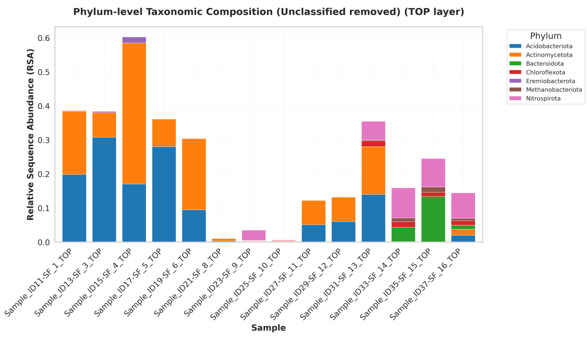
3.1 Layer-wise Distribution

This subsection compares the relative abundance of taxa between the TOP (61 bins) and BOTTOM (156 bins) soil layers, aggregated across all three locations.

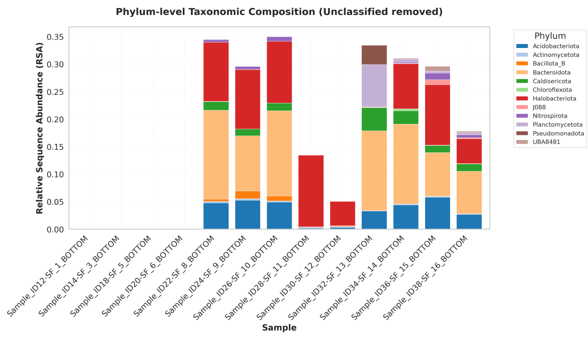
Table 1: Relative abundance (%) of phyla across TOP and BOTTOM soil layers in the Abisko soil metagenomics dataset. Values represent the percentage of bins in each layer assigned to the given phylum. Taxa are sorted by total abundance.

Phylum	**TOP (%)**	**BOTTOM (%)**
Acidobacteriota	26.23 (16)	24.36 (38)
Bacteroidota	13.11 (8)	20.51 (32)
Actinomycetota	44.26 (27)	1.28 (2)
Halobacteriota	0.00 (0)	16.67 (26)
Caldisericota	0.00 (0)	7.05 (11)
Desulfobacterota	0.00 (0)	5.13 (8)
Thermoproteota	0.00 (0)	5.13 (8)
Chloroflexota	1.64 (1)	4.49 (7)
Planctomycetota	0.00 (0)	5.13 (8)
Nitrospirota	6.56 (4)	1.92 (3)
Bacillota_B	0.00 (0)	1.92 (3)
Pseudomonadota	0.00 (0)	1.92 (3)
Verrucomicrobiota	0.00 (0)	1.28 (2)
Eremiobacterota	1.64 (1)	0.00 (0)
Methanobacteriota	1.64 (1)	0.00 (0)
SZUA-182	1.64 (1)	0.00 (0)
Desulfobacterota_E	0.00 (0)	0.64 (1)
J088	0.00 (0)	0.64 (1)
UBA8481	0.00 (0)	0.64 (1)

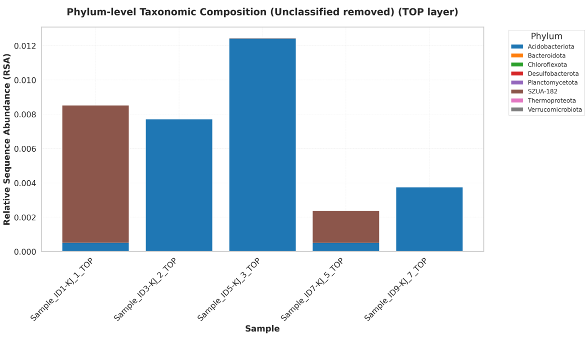
Acidobacteriota and Actinomycetota dominate the TOP layer with 26.23% and 44.26% relative abundance, respectively, while Actinomycetota collapses to 1.28% in the BOTTOM layer. In the BOTTOM layer, Acidobacteriota (24.36%), Bacteroidota (20.51%), and Halobacteriota (16.67%) constitute the most abundant phyla. Bacteroidota is present in both layers but shows greater abundance in the BOTTOM (20.51%) than in the TOP (13.11%). Halobacteriota, Caldisericota, Desulfobacterota, Thermoproteota, Planctomycetota, Bacillota_B, Pseudomonadota, Verrucomicrobiota, Desulfobacterota_E, J088, and UBA8481 are absent from the TOP layer. Conversely, Eremiobacterota, Methanobacteriota, and SZUA-182 are exclusive to the TOP layer. Five phyla (Acidobacteriota, Actinomycetota, Bacteroidota, Chloroflexota, Nitrospirota) occur in both layers.



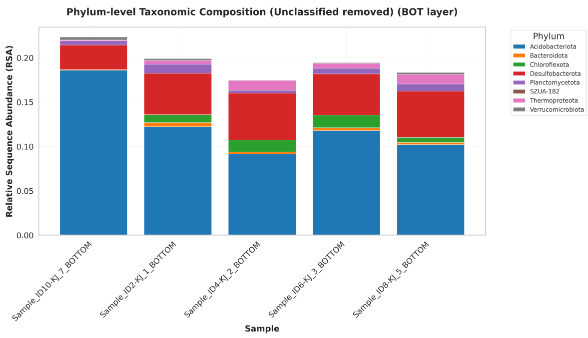
(a) SF TOP



(b) SF BOTTOM



(c) KJ TOP



(d) KJ BOTTOM

Figure 1: Phylum-level relative abundance distributions across sampling locations and depth layers. Each panel shows the percentage distribution of bins assigned to different phyla.

3.2 Cross-Layer Comparison

This subsection compares the bin counts of taxa across each site–layer combination. Note that the KF site contained no TOP-layer bins, and the KJ TOP layer contained only 2 bins.

Table 2: Phylum-level bin counts across each site and layer combination in the Abisko soil metagenomics dataset.

Phylum	**SF Top**	**SF Bot**	**KF Top**	**KF Bot**	**KJ Top**	**KJ Bot**
Acidobacteriota	15	16	0	0	1	22
Actinomycetota	27	1	0	1	0	0
Bacteroidota	8	29	0	1	0	2
Halobacteriota	0	25	0	1	0	0
Caldisericota	0	10	0	1	0	0
Desulfobacterota	0	0	0	0	0	8
Thermoproteota	0	0	0	0	0	8
Chloroflexota	1	1	0	2	0	4
Planctomycetota	0	4	0	1	0	3
Nitrospirota	4	3	0	0	0	0
Bacillota_B	0	3	0	0	0	0
Pseudomonadota	0	3	0	0	0	0
Verrucomicrobiota	0	0	0	0	0	2
Eremiobacterota	1	0	0	0	0	0
Methanobacteriota	1	0	0	0	0	0
SZUA-182	0	0	0	0	1	0
Desulfobacterota_E	0	0	0	1	0	0
J088	0	1	0	0	0	0
UBA8481	0	1	0	0	0	0

Acidobacteriota peaks in the SF Top (15 bins) and KJ Bottom (22 bins). Actinomycetota is highly concentrated in the SF Top (27 bins) but nearly absent elsewhere. Bacteroidota peaks in the SF Bottom (29 bins). Halobacteriota and Caldisericota are predominantly in the SF Bottom (25 and 10 bins, respectively). Desulfobacterota and Thermoproteota (8 bins each) are restricted to the KJ Bottom. Nitrospirota, Bacillota_B, Pseudomonadota, J088, UBA8481, Eremiobacterota, and Methanobacteriota are found only at SF.

3.3 Location-wise Distribution

This subsection aggregates bin counts at each location (SF: 158, KF: 8, KJ: 51) regardless of layer.

Table 3: Relative abundance (%) of phyla across the three sampling locations (SF, KF, KJ) in the Abisko soil metagenomics dataset.

Phylum	**SF (%)**	**KF (%)**	**KJ (%)**
Bacteroidota	23.42 (37)	12.50 (1)	3.92 (2)
Acidobacteriota	19.62 (31)	0.00 (0)	45.10 (23)
Actinomycetota	17.72 (28)	12.50 (1)	0.00 (0)
Halobacteriota	15.82 (25)	12.50 (1)	0.00 (0)
Caldisericota	6.33 (10)	12.50 (1)	0.00 (0)
Nitrospirota	4.43 (7)	0.00 (0)	0.00 (0)
Planctomycetota	2.53 (4)	12.50 (1)	5.88 (3)
Bacillota_B	1.90 (3)	0.00 (0)	0.00 (0)
Pseudomonadota	1.90 (3)	0.00 (0)	0.00 (0)
Chloroflexota	1.27 (2)	25.00 (2)	7.84 (4)
Desulfobacterota	0.00 (0)	0.00 (0)	15.69 (8)
Thermoproteota	0.00 (0)	0.00 (0)	15.69 (8)
Verrucomicrobiota	0.00 (0)	0.00 (0)	3.92 (2)
Eremiobacterota	0.63 (1)	0.00 (0)	0.00 (0)
Methanobacteriota	0.63 (1)	0.00 (0)	0.00 (0)
SZUA-182	0.00 (0)	0.00 (0)	1.96 (1)
Desulfobacterota_E	0.00 (0)	12.50 (1)	0.00 (0)
J088	0.63 (1)	0.00 (0)	0.00 (0)
UBA8481	0.63 (1)	0.00 (0)	0.00 (0)

SF is dominated by Bacteroidota (23.42%), Acidobacteriota (19.62%), Actinomycetota (17.72%), and Halobacteriota (15.82%). KJ is dominated by Acidobacteriota (45.10%), with Desulfobacterota and Thermoproteota (15.69% each) following. KF spans seven phyla (each represented by 1–2 bins), with Chloroflexota (25.00%) numerically the largest at this site. Only three phyla (Bacteroidota, Chloroflexota, Planctomycetota) appear in all three locations. Seven phyla are SF-exclusive, four KJ-exclusive, and one KF-exclusive (Desulfobacterota_E).

3.4 Cross-Location Comparison (Site and Layer)

This subsection combines location and layer dimensions. Bin counts are tabulated per site–layer combination, and the descriptive paragraph reports overall variation and presence/absence trends across all groups.

Table 4: Phylum bin counts across all six site–layer combinations in the Abisko soil metagenomics dataset.

Phylum	SF_Top	SF_Bot	KF_Top	KF_Bot	KJ_Top	KJ_Bot
Acidobacteriota	15	16	0	0	1	22
Bacteroidota	8	29	0	1	0	2
Actinomycetota	27	1	0	1	0	0
Halobacteriota	0	25	0	1	0	0
Caldisericota	0	10	0	1	0	0
Desulfobacterota	0	0	0	0	0	8
Thermoproteota	0	0	0	0	0	8
Chloroflexota	1	1	0	2	0	4
Planctomycetota	0	4	0	1	0	3
Nitrospirota	4	3	0	0	0	0
Bacillota_B	0	3	0	0	0	0
Pseudomonadota	0	3	0	0	0	0
Verrucomicrobiota	0	0	0	0	0	2
Eremiobacterota	1	0	0	0	0	0
Methanobacteriota	1	0	0	0	0	0
SZUA-182	0	0	0	0	1	0
Desulfobacterota_E	0	0	0	1	0	0
J088	0	1	0	0	0	0
UBA8481	0	1	0	0	0	0

The highest single bin counts are concentrated in SF_Bot (Bacteroidota: 29; Halobacteriota: 25; Acidobacteriota: 16), SF_Top (Actinomycetota: 27; Acidobacteriota: 15), and KJ_Bot (Acidobacteriota: 22). KF_Top contains zero phyla overall. Several phyla (Eremiobacterota, Methanobacteriota, J088, UBA8481, Desulfobacterota_E, SZUA-182) appear in only a single site–layer combination with one bin each. Acidobacteriota is the only phylum present in three site–layer groups (SF_Top, SF_Bot, KJ_Top, KJ_Bot).

3.5 Summary of Diversity per Group

Table 5: Bin counts and number of unique taxa per site–layer group in the Abisko soil metagenomics dataset.

Location	Layer	Bins	Phyla	Classes	Families	Genera
SF	TOP	59	7	9	9	14
SF	BOTTOM	99	12	13	15	15
KF	BOTTOM	8	7	8	8	7
KJ	TOP	2	2	2	1	1
KJ	BOTTOM	49	7	11	13	12

The SF Bottom group has the largest number of bins (99) and the highest taxonomic richness across all ranks (12 phyla, 13 classes, 15 families, 15 genera). The KJ Top group is the smallest (2 bins, 2 phyla, 2 classes, 1 family, 1 genus). The KF site contributed no TOP-layer bins.

4 Conclusion

This study applied nf-core mag pipeline to characterize microbial communities in Arctic permafrost soils from three sites in the Abisko region. Analysis of 64 samples comprising 1.9 billion paired-end reads resulted in 2,092 genome bins, with 158 high-quality MAGs taxonomically classified across at least 13 bacterial and archaeal phyla. The recovered community showed strong dominance of Acidobacteriota, alongside a substantial proportion of anaerobic taxa such as Desulfobacterota and Chloroflexota, indicating widespread anoxic microenvironments within soils. Archaeal representation, including methanogens, was notably elevated compared to typical soils, particularly in deeper layers, suggesting active methanogenic potential.

Clear stratification was observed between top and bottom soil layers, with surface communities dominated by aerobic and facultative heterotrophs, and deeper layers enriched in strict anaerobes such as sulfate-reducers and methanogens. Site-specific variation was also evident, with KJ showing the highest MAG recovery and strong enrichment of Acidobacteriota and Thermoproteota, while SF exhibited distinct signals including high Bacteroidota and unexpected Halobacteriota presence. KF yielded comparatively fewer high-quality MAGs, limiting interpretation. These findings highlight vertical hetergenity in permafrost microbial communities.

From a methodological perspective, the study demonstrates the effectiveness of the nf-core/mag pipeline implemented via Nextflow and containerized environments for large-scale metagenomic analysis. The combined use of multiple binning algorithms and GTDB-Tk classification enabled robust genome recovery and standardized taxonomy, while ensuring reproducibility through containerization and workflow management. Despite challenges such as fragmented assemblies and incomplete taxonomic resolution, the pipeline proved scalable and adaptable to complex environmental datasets.

Limitations include incomplete recovery of low-abundance taxa, unresolved classifications, and lack of functional validation. Future work should focus on genome annotation, experimental validation, and temporal sampling to better understand metabolic activity and ecosystem dynamics. Overall, this study provides a reproducible framework and foundational dataset for investigating Arctic soil microbiomes and their role in climate-relevant carbon cycling.

5 Appendix: Additional Taxonomic Results and Visualization

5.1 Layer-wise Distribution

Table 6: Relative abundance (%) of classes across TOP and BOTTOM soil layers in the Abisko soil metagenomics dataset.

Class	TOP (%)	BOTTOM (%)
Bog-38	0.00 (0)	16.67 (26)
Bacteroidia	13.11 (8)	16.67 (26)
Terriglobia	26.23 (16)	14.74 (23)
Actinomycetes	24.59 (15)	0.00 (0)
Thermoanaerobaculia	0.00 (0)	8.97 (14)
Caldisericia	0.00 (0)	7.05 (11)
Thermoleophilia	13.11 (8)	0.64 (1)
BSN033	0.00 (0)	5.13 (8)
Phycisphaerae	0.00 (0)	5.13 (8)
Bathyarchaeia	0.00 (0)	3.21 (5)
Acidimicrobiia	6.56 (4)	0.00 (0)
Anaerolineae	0.00 (0)	2.56 (4)
Thermodesulfovibrionia	6.56 (4)	1.92 (3)
DSM-12270	0.00 (0)	1.92 (3)
Gammaproteobacteria	0.00 (0)	1.92 (3)
Ignavibacteria	0.00 (0)	1.92 (3)
UBA10030	0.00 (0)	1.92 (3)
Nitrososphaeria	0.00 (0)	1.28 (2)
Verrucomicrobiae	0.00 (0)	1.28 (2)
Limnocylindria	1.64 (1)	1.28 (2)

Terriglobia (26.23%), Actinomycetes (24.59%), Bacteroidia (13.11%), and Thermoleophilia (13.11%) are the most abundant classes in the TOP layer. The BOTTOM layer is dominated by Bog-38 (16.67%), Bacteroidia (16.67%), and Terriglobia (14.74%). Actinomycetes, Acidimicrobiia, Eremiobacteria, Methanobacteria, and SZUA-182 are exclusive to the TOP. Bog-38, Thermoanaerobaculia, Caldisericia, BSN033, Phycisphaerae, Bathyarchaeia, Anaerolineae, DSM-12270, Gammaproteobacteria, Ignavibacteria, UBA10030, Nitrososphaeria, Verrucomicrobiae, Methanomethylicia, Aminicenantia, Deferrimicrobia, Dehalococcoidia, and Geothermincolia are exclusive to the BOTTOM. Five classes (Bacteroidia, Limnocylindria, Terriglobia, Thermodesulfovibrionia, Thermoleophilia) appear in both layers.

Table 7: Relative abundance (%) of orders across TOP and BOTTOM soil layers in the Abisko soil metagenomics dataset.

Order	**TOP (%)**	**BOTTOM (%)**
Bacteroidales	13.11 (8)	16.67 (26)
Bog-38	0.00 (0)	16.67 (26)
Streptosporangiales	24.59 (15)	0.00 (0)
Acidoferrales	18.03 (11)	3.85 (6)
Cryosericales	0.00 (0)	7.05 (11)
Thermoanaerobaculales	0.00 (0)	7.05 (11)
UBA7540	0.00 (0)	6.41 (10)
Solirubrobacterales	13.11 (8)	0.00 (0)
BSN033	0.00 (0)	5.13 (8)
Sedimentisphaerales	0.00 (0)	5.13 (8)
Bathyarchaeales	0.00 (0)	3.21 (5)
Acidimicrobiales	6.56 (4)	0.00 (0)
Anaerolineales	0.00 (0)	2.56 (4)
Terriglobales	8.20 (5)	1.92 (3)
Thermodesulfovibrionales	6.56 (4)	1.92 (3)
Bryobacterales	0.00 (0)	2.56 (4)
Steroidobacterales	0.00 (0)	1.92 (3)
Thermacetogeniales	0.00 (0)	1.92 (3)
UBA10030	0.00 (0)	1.92 (3)
UBA5066	0.00 (0)	1.92 (3)

Streptosporangiales (24.59%), Acidoferrales (18.03%), Bacteroidales (13.11%), and Solirubrobacterales (13.11%) are the leading orders in the TOP layer. In the BOTTOM, Bacteroidales (16.67%) and Bog-38 (16.67%) are co-dominant, followed by Cryosericales (7.05%) and Thermoanaerobaculales (7.05%). Five orders are shared across layers (Acidoferrales, Bacteroidales, Limnocylindrales, Terriglobales, Thermodesulfovibrionales); four are exclusive to the TOP (Acidimicrobiales, Methanobacteriales, Solirubrobacterales, Streptosporangiales); and 22 orders are exclusive to the BOTTOM layer.

Table 8: Relative abundance (%) of families across TOP and BOTTOM soil layers in the Abisko soil metagenomics dataset.

Family	TOP (%)	BOTTOM (%)
Bog-38	0.00 (0)	16.67 (26)
VadinHA17	13.11 (8)	14.74 (23)
UBA7541	18.03 (11)	3.85 (6)
Streptosporangiaceae	24.59 (15)	0.00 (0)
Cryosericaceae	0.00 (0)	7.05 (11)
Thermoanaerobaculaceae	0.00 (0)	7.05 (11)
UBA7540	0.00 (0)	6.41 (10)
Solirubrobacteraceae	13.11 (8)	0.00 (0)
UBA1163	0.00 (0)	5.13 (8)
SG8-4	0.00 (0)	3.21 (5)
SbA1	8.20 (5)	0.00 (0)
UBA233	0.00 (0)	3.21 (5)
Bryobacteraceae	0.00 (0)	2.56 (4)
Dissulfurispiraceae	6.56 (4)	1.92 (3)
RAAP-2	6.56 (4)	0.00 (0)
Villigracilaceae	0.00 (0)	2.56 (4)
Acidobacteriaceae	0.00 (0)	1.92 (3)
FEN-979	0.00 (0)	1.92 (3)
Steroidobacteraceae	0.00 (0)	1.92 (3)
Thermacetogeniaceae	0.00 (0)	1.92 (3)

Streptosporangiaceae (24.59%), UBA7541 (18.03%), VadinHA17 (13.11%), and Solirubrobacteraceae (13.11%) lead the TOP layer. The BOTTOM is dominated by Bog-38 (16.67%) and VadinHA17 (14.74%), followed by Cryosericaceae (7.05%) and Thermoanaerobaculaceae (7.05%). Four families (CSP1-4, Dissulfurispiraceae, UBA7541, VadinHA17) co-occur in both layers. Five families are TOP-exclusive (Methanobacteriaceae, RAAP-2, SbA1, Solirubrobacteraceae, Streptosporangiaceae), and 24 are BOTTOM-exclusive.

Table 9: Relative abundance (%) of genera across TOP and BOTTOM soil layers in the Abisko soil metagenomics dataset.

Genus	TOP (%)	BOTTOM (%)
LD21	13.11 (8)	14.74 (23)
Bog-38	0.00 (0)	16.67 (26)
Cryosericum	0.00 (0)	6.41 (10)
RBG-13-68-16	0.00 (0)	5.77 (9)
UBA1163	0.00 (0)	5.13 (8)
UBA7540	0.00 (0)	5.13 (8)
Palsa-295	13.11 (8)	0.00 (0)
Palsa-465	13.11 (8)	0.00 (0)
Palsa-506	9.84 (6)	0.00 (0)
Acidoferrum	0.00 (0)	3.85 (6)
DRUK01	0.00 (0)	3.21 (5)
Fen-1308	6.56 (4)	1.92 (3)
QIAJ01	8.20 (5)	0.00 (0)
RAAP-2	6.56 (4)	0.00 (0)
UBA877	0.00 (0)	2.56 (4)
Bog-532	4.92 (3)	0.00 (0)
Palsa-147	4.92 (3)	0.00 (0)
Bog-1198	0.00 (0)	1.92 (3)
Fen-1298	0.00 (0)	1.92 (3)
Fen-1301	0.00 (0)	1.92 (3)

LD21 (13.11%), Palsa-295 (13.11%), Palsa-465 (13.11%), and Palsa-506 (9.84%) are the leading genera in the TOP layer. In the BOTTOM, Bog-38 (16.67%) and LD21 (14.74%) dominate, with Cryosericum (6.41%) and RBG-13-68-16 (5.77%) also notable. Only three genera occur in both layers (CTSoil-043, Fen-1308, LD21). Eleven genera are TOP-exclusive, while 25 genera are BOTTOM-exclusive.

Table 10: Relative abundance (%) of species across TOP and BOTTOM soil layers in the Abisko soil metagenomics dataset.

Species	TOP (%)	BOTTOM (%)
Bog-38 sp003162175	0.00 (0)	5.77 (9)
LD21 sp003141415	0.00 (0)	5.13 (8)
LD21 sp003141495	1.64 (1)	5.13 (8)
RBG-13-68-16 sp003133645	0.00 (0)	5.13 (8)
Fen-1308 sp003170655	6.56 (4)	0.00 (0)
Palsa-295 sp003153585	6.56 (4)	0.00 (0)
Palsa-465 sp003165655	6.56 (4)	0.00 (0)
Palsa-506 sp003168215	6.56 (4)	0.00 (0)
Palsa-295 sp003131985	4.92 (3)	0.00 (0)
Bog-38 sp003170935	0.00 (0)	1.92 (3)
FEN-979 sp003142515	0.00 (0)	1.92 (3)
Fen-1298 sp003142575	0.00 (0)	1.92 (3)
Fen-183 sp003152095	0.00 (0)	1.92 (3)
LD21 sp003141895	0.00 (0)	1.92 (3)
UBA12454 sp003142595	0.00 (0)	1.92 (3)
Fen-1467 sp903940425	0.00 (0)	1.28 (2)
Cryosericum hinesii	0.00 (0)	0.64 (1)
Deferrimicrobium sp937863105	0.00 (0)	0.64 (1)
Occallatibacter sp003133425	0.00 (0)	0.64 (1)
Palsa-1188 sp003132605	0.00 (0)	0.64 (1)

Only one species, *LD21 sp003141495*, is shared between the layers (TOP: 1.64%; BOTTOM: 5.13%). The TOP layer is characterized by *Fen-1308 sp003170655*, *Palsa-295 sp003153585*, *Palsa-465 sp003165655*, and *Palsa-506 sp003168215* (each at 6.56%). The BOTTOM layer is dominated by *Bog-38 sp003162175* (5.77%), and three species at 5.13% each (*LD21 sp003141415*, *LD21 sp003141495*, *RBG-13-68-16 sp003133645*). Five species are TOP-exclusive; 14 are BOTTOM-exclusive.

5.2 Cross-Layer Distribution

Table 11: Class-level bin counts across each site and layer combination in the Abisko soil metagenomics dataset.

Class	SF Top	SF Bot	KF Top	KF Bot	KJ Top	KJ Bot
Bog-38	0	25	0	1	0	0
Bacteroidia	8	23	0	1	0	2
Terriglobia	15	4	0	0	1	19
Actinomycetes	15	0	0	0	0	0
Thermoanaerobaculia	0	12	0	0	0	2
Caldisericia	0	10	0	1	0	0
Thermoleophilia	8	1	0	0	0	0
BSN033	0	0	0	0	0	8
Phycisphaerae	0	4	0	1	0	3
Bathyarchaeia	0	0	0	0	0	5
Anaerolineae	0	0	0	0	0	4
Acidimicrobiia	4	0	0	0	0	0
Thermodesulfovibrionia	4	3	0	0	0	0
DSM-12270	0	3	0	0	0	0
Gammaproteobacteria	0	3	0	0	0	0
Ignavibacteria	0	3	0	0	0	0
UBA10030	0	3	0	0	0	0
Nitrososphaeria	0	0	0	0	0	2
Verrucomicrobiae	0	0	0	0	0	2
Limnocylindria	1	1	0	1	0	0

Bog-38 dominates the SF Bottom (25 bins) with a single bin in the KF Bottom. Terriglobia is the dominant class in the KJ Bottom (19 bins) and is also abundant in the SF Top (15 bins). Actinomycetes (15 bins) and Thermoleophilia (8 bins) appear exclusively in the SF Top. BSN033, Bathyarchaeia, Anaerolineae, Nitrososphaeria, and Verrucomicrobiae are KJ-Bottom exclusives. KF Top and KJ Top contribute very few classes overall (0 and 2, respectively).

Table 12: Order-level bin counts across each site and layer combination in the Abisko soil metagenomics dataset.

Order	**SF Top**	**SF Bot**	**KF Top**	**KF Bot**	**KJ Top**	**KJ Bot**
Bacteroidales	8	23	0	1	0	2
Bog-38	0	25	0	1	0	0
Streptosporangiales	15	0	0	0	0	0
Acidoferrales	11	0	0	0	0	6
UBA7540	0	0	0	0	0	10
Cryosericales	0	10	0	1	0	0
Thermoanaerobaculales	0	9	0	0	0	2
Solirubrobacterales	8	0	0	0	0	0
BSN033	0	0	0	0	0	8
Sedimentisphaerales	0	4	0	1	0	3
Bathyarchaeales	0	0	0	0	0	5
Anaerolineales	0	0	0	0	0	4
Terriglobales	4	3	0	0	1	0
Thermodesulfovibrionales	4	3	0	0	0	0
Acidimicrobiales	4	0	0	0	0	0
Bryobacterales	0	1	0	0	0	3
Steroidobacterales	0	3	0	0	0	0
Thermacetogeniales	0	3	0	0	0	0
UBA5066	0	3	0	0	0	0
UBA10030	0	3	0	0	0	0

Bacteroidales (23) and Bog-38 (25) dominate the SF Bottom. Streptosporangiales (15), Acidoferrales (11), and Solirubrobacterales (8) dominate the SF Top. UBA7540 (10) and BSN033 (8) are KJ-Bottom exclusives. Cryosericales is essentially a SF Bottom (10) order with minor presence at KF Bottom (1). The KF and KJ Top layers contribute almost no orders (0 and 1, respectively).

Table 13: Family-level bin counts across each site and layer combination in the Abisko soil metagenomics dataset.

Family	SF Top	SF Bot	KF Top	KF Bot	KJ Top	KJ Bot
VadinHA17	8	20	0	1	0	2
Bog-38	0	25	0	1	0	0
Streptosporangiaceae	15	0	0	0	0	0
UBA7541	11	0	0	0	0	6
UBA7540	0	0	0	0	0	10
Cryosericaceae	0	10	0	1	0	0
Thermoanaerobaculaceae	0	9	0	0	0	2
UBA1163	0	0	0	0	0	8
Solirubrobacteraceae	8	0	0	0	0	0
SbA1	4	0	0	0	1	0
UBA233	0	0	0	0	0	5
Dissulfurispiraceae	4	3	0	0	0	0
RAAP-2	4	0	0	0	0	0
Villigracilaceae	0	0	0	0	0	4
Bryobacteraceae	0	1	0	0	0	3
SG8-4	0	1	0	1	0	3
Acidobacteriaceae	0	3	0	0	0	0
FEN-979	0	3	0	0	0	0
Steroidobacteraceae	0	3	0	0	0	0
Thermacetogeniaceae	0	3	0	0	0	0

VadinHA17 is the most widely distributed family across site–layer combinations (SF Top: 8, SF Bottom: 20, KF Bottom: 1, KJ Bottom: 2). Bog-38 again dominates the SF Bottom (25 bins). Streptosporangiaceae (15) and UBA7541 (11) lead the SF Top. UBA7540 (10) and UBA1163 (8) are restricted to the KJ Bottom. SG8-4 is the only family present in three site–layer groups beyond SF Top (SF Bottom: 1, KF Bottom: 1, KJ Bottom: 3).

Table 14: Genus-level bin counts across each site and layer combination in the Abisko soil metagenomics dataset.

Genus	SF Top	SF Bot	KF Top	KF Bot	KJ Top	KJ Bot
LD21	8	20	0	1	0	2
Bog-38	0	25	0	1	0	0
Cryosericum	0	9	0	1	0	0
RBG-13-68-16	0	9	0	0	0	0
UBA1163	0	0	0	0	0	8
UBA7540	0	0	0	0	0	8
Palsa-295	8	0	0	0	0	0
Palsa-465	8	0	0	0	0	0
Palsa-506	6	0	0	0	0	0
Acidoferrum	0	0	0	0	0	6
DRUK01	0	0	0	0	0	5
QIAJ01	4	0	0	0	1	0
RAAP-2	4	0	0	0	0	0
Fen-1308	4	3	0	0	0	0
UBA877	0	0	0	0	0	4
Bog-532	3	0	0	0	0	0
Palsa-147	3	0	0	0	0	0
Bog-1198	0	3	0	0	0	0
Fen-1298	0	3	0	0	0	0
Fen-1301	0	3	0	0	0	0

LD21 is the most broadly distributed genus, with the highest count in the SF Bottom (20) and noticeable counts in the SF Top (8), KJ Bottom (2), and KF Bottom (1). Bog-38 (25) and Cryosericum (9) are largely SF Bottom genera, while RBG-13-68-16 (9) is SF-Bottom exclusive. The SF Top is characterized by Palsa-295 (8), Palsa-465 (8), Palsa-506 (6), QIAJ01 (4), and RAAP-2 (4). The KJ Bottom holds the highest counts of UBA1163 (8), UBA7540 (8), Acidoferrum (6), DRUK01 (5), and UBA877 (4).

Table 15: Species-level bin counts across each site and layer combination in the Abisko soil metagenomics dataset.

Species	**SF Top**	**SF Bot**	**KF Top**	**KF Bot**	**KJ Top**	**KJ Bot**
Bog-38 sp003162175	0	9	0	0	0	0
LD21 sp003141415	0	8	0	0	0	0
LD21 sp003141495	1	8	0	0	0	0
RBG-13-68-16 sp003133645	0	8	0	0	0	0
Fen-1308 sp003170655	4	0	0	0	0	0
Palsa-295 sp003153585	4	0	0	0	0	0
Palsa-465 sp003165655	4	0	0	0	0	0
Palsa-506 sp003168215	4	0	0	0	0	0
Palsa-295 sp003131985	3	0	0	0	0	0
Bog-38 sp003170935	0	3	0	0	0	0
FEN-979 sp003142515	0	3	0	0	0	0
Fen-1298 sp003142575	0	3	0	0	0	0
Fen-183 sp003152095	0	3	0	0	0	0
LD21 sp003141895	0	3	0	0	0	0
UBA12454 sp003142595	0	3	0	0	0	0
Fen-1467 sp903940425	0	0	0	0	0	2
Cryosericum hinesii	0	0	0	1	0	0
Deferrimicrobium sp937863105	0	0	0	1	0	0
Occallatibacter sp003133425	0	1	0	0	0	0
Palsa-1188 sp003132605	0	1	0	0	0	0

The SF Bottom holds the largest counts of *Bog-38 sp003162175* (9), *LD21 sp003141415* (8), *LD21 sp003141495* (8), and *RBG-13-68-16 sp003133645* (8). The SF Top is led by four species at 4 bins each (*Fen-1308 sp003170655*, *Palsa-295 sp003153585*, *Palsa-465 sp003165655*, *Palsa-506 sp003168215*). Only one species is shared between two site–layer groups (*LD21 sp003141495*: 1 in SF Top, 8 in SF Bottom). The KF Bottom contains two singleton species (*Cryosericum hinesii*, *Deferrimicrobium sp937863105*). The KJ Bottom contains a single species (*Fen-1467 sp903940425*, 2 bins).

5.3 Location-wise Distribution

Table 16: Relative abundance (%) of classes across SF, KF, and KJ in the Abisko soil metagenomics dataset.

Class	SF (%)	KF (%)	KJ (%)
Bacteroidia	19.62 (31)	12.50 (1)	3.92 (2)
Bog-38	15.82 (25)	12.50 (1)	0.00 (0)
Terriglobia	12.03 (19)	0.00 (0)	39.22 (20)
Actinomycetes	9.49 (15)	0.00 (0)	0.00 (0)
Thermoanaerobaculia	7.59 (12)	0.00 (0)	3.92 (2)
Caldisericia	6.33 (10)	12.50 (1)	0.00 (0)
Thermoleophilia	5.70 (9)	0.00 (0)	0.00 (0)
BSN033	0.00 (0)	0.00 (0)	15.69 (8)
Thermodesulfovibrionia	4.43 (7)	0.00 (0)	0.00 (0)
Bathyarchaeia	0.00 (0)	0.00 (0)	9.80 (5)
Phycisphaerae	2.53 (4)	12.50 (1)	5.88 (3)
Anaerolineae	0.00 (0)	0.00 (0)	7.84 (4)
Acidimicrobiia	2.53 (4)	0.00 (0)	0.00 (0)
DSM-12270	1.90 (3)	0.00 (0)	0.00 (0)
Gammaproteobacteria	1.90 (3)	0.00 (0)	0.00 (0)
Ignavibacteria	1.90 (3)	0.00 (0)	0.00 (0)
UBA10030	1.90 (3)	0.00 (0)	0.00 (0)
Nitrososphaeria	0.00 (0)	0.00 (0)	3.92 (2)
Verrucomicrobiae	0.00 (0)	0.00 (0)	3.92 (2)
Limnocylindria	1.27 (2)	12.50 (1)	0.00 (0)

Bacteroidia (19.62%), Bog-38 (15.82%), and Terriglobia (12.03%) are the most abundant classes in SF. Terriglobia dominates KJ (39.22%), followed by BSN033 (15.69%) and Bathyarchaeia (9.80%). KF spreads its eight classes uniformly (each at 12.50%). Only Bacteroidia and Phycisphaerae occur across all three sites. Ten classes are exclusive to SF, eight to KJ, and three to KF.

Table 17: Relative abundance (%) of orders across SF, KF, and KJ in the Abisko soil metagenomics dataset.

Order	SF (%)	KF (%)	KJ (%)
Bacteroidales	19.62 (31)	12.50 (1)	3.92 (2)
Bog-38	15.82 (25)	12.50 (1)	0.00 (0)
Streptosporangiales	9.49 (15)	0.00 (0)	0.00 (0)
Acidoferrales	6.96 (11)	0.00 (0)	11.76 (6)
UBA7540	0.00 (0)	0.00 (0)	19.61 (10)
Cryosericales	6.33 (10)	12.50 (1)	0.00 (0)
Thermoanaerobaculales	5.70 (9)	0.00 (0)	3.92 (2)
Solirubrobacterales	5.06 (8)	0.00 (0)	0.00 (0)
BSN033	0.00 (0)	0.00 (0)	15.69 (8)
Thermodesulfovibrionales	4.43 (7)	0.00 (0)	0.00 (0)
Terriglobales	4.43 (7)	0.00 (0)	1.96 (1)
Bathyarchaeales	0.00 (0)	0.00 (0)	9.80 (5)
Sedimentisphaerales	2.53 (4)	12.50 (1)	5.88 (3)
Anaerolineales	0.00 (0)	0.00 (0)	7.84 (4)
Acidimicrobiales	2.53 (4)	0.00 (0)	0.00 (0)
Bryobacterales	0.63 (1)	0.00 (0)	5.88 (3)
Steroidobacterales	1.90 (3)	0.00 (0)	0.00 (0)
Thermacetogeniales	1.90 (3)	0.00 (0)	0.00 (0)
UBA5066	1.90 (3)	0.00 (0)	0.00 (0)
UBA10030	1.90 (3)	0.00 (0)	0.00 (0)

Bacteroidales (19.62%) and Bog-38 (15.82%) are the most abundant orders in SF, followed by Streptosporangiales (9.49%). KJ is dominated by UBA7540 (19.61%), BSN033 (15.69%), and Acidoferrales (11.76%). KF spreads its orders uniformly across 8 entries. Only Bacteroidales and Sedimentisphaerales are common to all three sites. Eleven orders are SF-unique, eight KJ-unique, and three KF-unique.

Table 18: Relative abundance (%) of families across SF, KF, and KJ in the Abisko soil metagenomics dataset.

Family	SF (%)	KF (%)	KJ (%)
VadinHA17	17.72 (28)	12.50 (1)	3.92 (2)
Bog-38	15.82 (25)	12.50 (1)	0.00 (0)
Streptosporangiaceae	9.49 (15)	0.00 (0)	0.00 (0)
UBA7541	6.96 (11)	0.00 (0)	11.76 (6)
UBA7540	0.00 (0)	0.00 (0)	19.61 (10)
Cryosericaceae	6.33 (10)	12.50 (1)	0.00 (0)
Thermoanaerobaculaceae	5.70 (9)	0.00 (0)	3.92 (2)
Solirubrobacteraceae	5.06 (8)	0.00 (0)	0.00 (0)
UBA1163	0.00 (0)	0.00 (0)	15.69 (8)
Dissulfurispiraceae	4.43 (7)	0.00 (0)	0.00 (0)
SbA1	2.53 (4)	0.00 (0)	1.96 (1)
RAAP-2	2.53 (4)	0.00 (0)	0.00 (0)
UBA233	0.00 (0)	0.00 (0)	9.80 (5)
Villigracilaceae	0.00 (0)	0.00 (0)	7.84 (4)
Bryobacteraceae	0.63 (1)	0.00 (0)	5.88 (3)
SG8-4	0.63 (1)	12.50 (1)	5.88 (3)
Acidobacteriaceae	1.90 (3)	0.00 (0)	0.00 (0)
FEN-979	1.90 (3)	0.00 (0)	0.00 (0)
Steroidobacteraceae	1.90 (3)	0.00 (0)	0.00 (0)
Thermacetogeniaceae	1.90 (3)	0.00 (0)	0.00 (0)

VadinHA17 (17.72%) and Bog-38 (15.82%) dominate SF families, while UBA7540 (19.61%) and UBA1163 (15.69%) dominate KJ. SG8-4 and VadinHA17 are the only families common to all three locations. KF has 8 unique families each at 12.50%. SF has 13 unique families, and KJ has 8 unique families.

Table 19: Relative abundance (%) of genera across SF, KF, and KJ in the Abisko soil metagenomics dataset.

Genus	SF (%)	KF (%)	KJ (%)
LD21	17.72 (28)	12.50 (1)	3.92 (2)
Bog-38	15.82 (25)	12.50 (1)	0.00 (0)
Cryosericum	5.70 (9)	12.50 (1)	0.00 (0)
RBG-13-68-16	5.70 (9)	0.00 (0)	0.00 (0)
UBA1163	0.00 (0)	0.00 (0)	15.69 (8)
UBA7540	0.00 (0)	0.00 (0)	15.69 (8)
Palsa-295	5.06 (8)	0.00 (0)	0.00 (0)
Palsa-465	5.06 (8)	0.00 (0)	0.00 (0)
Palsa-506	3.80 (6)	0.00 (0)	0.00 (0)
Acidoferrum	0.00 (0)	0.00 (0)	11.76 (6)
Fen-1308	4.43 (7)	0.00 (0)	0.00 (0)
DRUK01	0.00 (0)	0.00 (0)	9.80 (5)
QIAJ01	2.53 (4)	0.00 (0)	1.96 (1)
RAAP-2	2.53 (4)	0.00 (0)	0.00 (0)
UBA877	0.00 (0)	0.00 (0)	7.84 (4)
Fen-1362	0.63 (1)	12.50 (1)	5.88 (3)
Bog-1198	1.90 (3)	0.00 (0)	0.00 (0)
Bog-532	1.90 (3)	0.00 (0)	0.00 (0)
Fen-1298	1.90 (3)	0.00 (0)	0.00 (0)
Fen-1301	1.90 (3)	0.00 (0)	0.00 (0)

LD21 (17.72%) and Bog-38 (15.82%) are the leading genera in SF. In KJ, UBA1163 (15.69%), UBA7540 (15.69%), and Acidoferrum (11.76%) are the most abundant. Fen-1362 and LD21 are the only two genera shared among all three locations. SF has 21 unique genera, KJ has 10 unique genera, and KF has 2 unique genera.

Table 20: Relative abundance (%) of species across SF, KF, and KJ in the Abisko soil metagenomics dataset.

Species	SF (%)	KF (%)	KJ (%)
Bog-38 sp003162175	5.70 (9)	0.00 (0)	0.00 (0)
LD21 sp003141415	5.06 (8)	0.00 (0)	0.00 (0)
LD21 sp003141495	5.70 (9)	0.00 (0)	0.00 (0)
RBG-13-68-16 sp003133645	5.06 (8)	0.00 (0)	0.00 (0)
Fen-1308 sp003170655	2.53 (4)	0.00 (0)	0.00 (0)
Palsa-295 sp003153585	2.53 (4)	0.00 (0)	0.00 (0)
Palsa-465 sp003165655	2.53 (4)	0.00 (0)	0.00 (0)
Palsa-506 sp003168215	2.53 (4)	0.00 (0)	0.00 (0)
Bog-38 sp003170935	1.90 (3)	0.00 (0)	0.00 (0)
FEN-979 sp003142515	1.90 (3)	0.00 (0)	0.00 (0)
Fen-1298 sp003142575	1.90 (3)	0.00 (0)	0.00 (0)
Fen-183 sp003152095	1.90 (3)	0.00 (0)	0.00 (0)
LD21 sp003141895	1.90 (3)	0.00 (0)	0.00 (0)
Palsa-295 sp003131985	1.90 (3)	0.00 (0)	0.00 (0)
UBA12454 sp003142595	1.90 (3)	0.00 (0)	0.00 (0)
Fen-1467 sp903940425	0.00 (0)	0.00 (0)	3.92 (2)
Cryosericum hinesii	0.00 (0)	12.50 (1)	0.00 (0)
Deferrimicrobium sp937863105	0.00 (0)	12.50 (1)	0.00 (0)
Occallatibacter sp003133425	0.63 (1)	0.00 (0)	0.00 (0)
Palsa-1188 sp003132605	0.63 (1)	0.00 (0)	0.00 (0)

No species is shared across all three locations. SF retains 17 unique species, led by *Bog-38 sp003162175* (5.70%) and *LD21 sp003141495* (5.70%). KF contains 2 unique species (*Cryosericum hinesii*, *Deferrimicrobium sp937863105*; both 12.50%). KJ contains 1 unique species (*Fen-1467 sp903940425*, 3.92%).

5.4 Cross-Location Comparison (Site and Layer)

Table 21: Class bin counts across all six site–layer combinations in the Abisko soil metagenomics dataset.

Class	SF_Top	SF_Bot	KF_Top	KF_Bot	KJ_Top	KJ_Bot
Bog-38	0	25	0	1	0	0
Bacteroidia	8	23	0	1	0	2
Terriglobia	15	4	0	0	1	19
Actinomycetes	15	0	0	0	0	0
Thermoanaerobaculia	0	12	0	0	0	2
Caldisericia	0	10	0	1	0	0
Thermoleophilia	8	1	0	0	0	0
BSN033	0	0	0	0	0	8
Bathyarchaeia	0	0	0	0	0	5
Anaerolineae	0	0	0	0	0	4
Thermodesulfovibrionia	4	3	0	0	0	0
Phycisphaerae	0	4	0	1	0	3
Acidimicrobiia	4	0	0	0	0	0
DSM-12270	0	3	0	0	0	0
Gammaproteobacteria	0	3	0	0	0	0
Ignavibacteria	0	3	0	0	0	0
UBA10030	0	3	0	0	0	0
Nitrososphaeria	0	0	0	0	0	2
Verrucomicrobiae	0	0	0	0	0	2
Limnocylindria	1	1	0	1	0	0

Maximum class counts: Bog-38 (25 in SF_Bot), Bacteroidia (23 in SF_Bot), Terriglobia (19 in KJ_Bot, also 15 in SF_Top), Actinomycetes (15 in SF_Top), Thermoanaerobaculia (12 in SF_Bot), Caldisericia (10 in SF_Bot), and BSN033 (8 in KJ_Bot). KF_Top is empty across all classes. KJ_Top contains only 2 classes (SZUA-182, Terriglobia), each with 1 bin. Phycisphaerae appears in three site–layer combinations (SF_Bot, KF_Bot, KJ_Bot); Limnocylindria also spans three combinations (SF_Top, SF_Bot, KF_Bot).

Table 22: Order bin counts across all six site–layer combinations in the Abisko soil metagenomics dataset.

Order	SF_Top	SF_Bot	KF_Top	KF_Bot	KJ_Top	KJ_Bot
Bog-38	0	25	0	1	0	0
Bacteroidales	8	23	0	1	0	2
Streptosporangiales	15	0	0	0	0	0
Acidoferrales	11	0	0	0	0	6
UBA7540	0	0	0	0	0	10
Cryosericales	0	10	0	1	0	0
Thermoanaerobaculales	0	9	0	0	0	2
Solirubrobacterales	8	0	0	0	0	0
BSN033	0	0	0	0	0	8
Sedimentisphaerales	0	4	0	1	0	3
Bathyarchaeales	0	0	0	0	0	5
Anaerolineales	0	0	0	0	0	4
Terriglobales	4	3	0	0	1	0
Thermodesulfovibrionales	4	3	0	0	0	0
Bryobacterales	0	1	0	0	0	3
Acidimicrobiales	4	0	0	0	0	0
Steroidobacterales	0	3	0	0	0	0
Thermacetogeniales	0	3	0	0	0	0
UBA5066	0	3	0	0	0	0
UBA10030	0	3	0	0	0	0

Bog-38 (25) and Bacteroidales (23) reach their highest counts in SF_Bot. Streptosporangiales (15) is the most abundant order in SF_Top. UBA7540 (10) and BSN033 (8) reach their highest counts in KJ_Bot. KF_Top contains no orders. The orders Bacteroidales and Sedimentisphaerales span four site–layer combinations each, representing the broadest distribution at this rank. Most orders are present in only one or two site–layer combinations.

Table 23: Family bin counts across all six site–layer combinations in the Abisko soil metagenomics dataset.

Family	**SF_Top**	**SF_Bot**	**KF_Top**	**KF_Bot**	**KJ_Top**	**KJ_Bot**
VadinHA17	8	20	0	1	0	2
Bog-38	0	25	0	1	0	0
Streptosporangiaceae	15	0	0	0	0	0
UBA7541	11	0	0	0	0	6
UBA7540	0	0	0	0	0	10
Cryosericaceae	0	10	0	1	0	0
Thermoanaerobaculaceae	0	9	0	0	0	2
UBA1163	0	0	0	0	0	8
Solirubrobacteraceae	8	0	0	0	0	0
SbA1	4	0	0	0	1	0
UBA233	0	0	0	0	0	5
Dissulfurispiraceae	4	3	0	0	0	0
RAAP-2	4	0	0	0	0	0
Villigracilaceae	0	0	0	0	0	4
Bryobacteraceae	0	1	0	0	0	3
SG8-4	0	1	0	1	0	3
Acidobacteriaceae	0	3	0	0	0	0
FEN-979	0	3	0	0	0	0
Steroidobacteraceae	0	3	0	0	0	0
Thermacetogeniaceae	0	3	0	0	0	0

VadinHA17 spans four site–layer combinations (SF_Top: 8, SF_Bot: 20, KF_Bot: 1, KJ_Bot: 2), with its maximum in SF_Bot. Bog-38 (25 in SF_Bot), Streptosporangiaceae (15 in SF_Top), UBA7541 (11 in SF_Top), UBA7540 (10 in KJ_Bot), Cryosericaceae (10 in SF_Bot), and Thermoanaerobaculaceae (9 in SF_Bot) are the highest single-cell counts. SG8-4 is observed in three combinations (SF_Bot, KF_Bot, KJ_Bot). KF_Top and KJ_Top contribute almost no families (0 and 1, respectively).

Table 24: Genus bin counts across all six site–layer combinations in the Abisko soil metagenomics dataset.

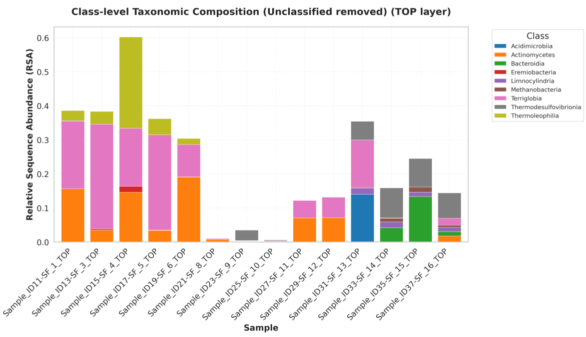
Genus	**SF_Top**	**SF_Bot**	**KF_Top**	**KF_Bot**	**KJ_Top**	**KJ_Bot**
LD21	8	20	0	1	0	2
Bog-38	0	25	0	1	0	0
Cryosericum	0	9	0	1	0	0
RBG-13-68-16	0	9	0	0	0	0
UBA1163	0	0	0	0	0	8
UBA7540	0	0	0	0	0	8
Palsa-295	8	0	0	0	0	0
Palsa-465	8	0	0	0	0	0
Palsa-506	6	0	0	0	0	0
Acidoferrum	0	0	0	0	0	6
DRUK01	0	0	0	0	0	5
Fen-1308	4	3	0	0	0	0
QIAJ01	4	0	0	0	1	0
RAAP-2	4	0	0	0	0	0
UBA877	0	0	0	0	0	4
Bog-532	3	0	0	0	0	0
Palsa-147	3	0	0	0	0	0
Bog-1198	0	3	0	0	0	0
Fen-1298	0	3	0	0	0	0
Fen-1301	0	3	0	0	0	0

LD21 spans four site–layer combinations (SF_Top: 8, SF_Bot: 20, KF_Bot: 1, KJ_Bot: 2). Bog-38 (25), Cryosericum (9), and RBG-13-68-16 (9) reach their highest counts in SF_Bot. Palsa-295 (8), Palsa-465 (8), and Palsa-506 (6) are restricted to SF_Top. UBA1163 (8), UBA7540 (8), Acidoferrum (6), and DRUK01 (5) are restricted to KJ_Bot. KF_Top has 0 genera; KJ_Top has only 1 (QIAJ01: 1 bin).

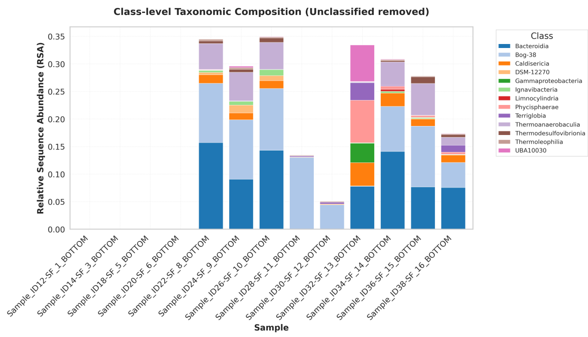
Table 25: Species bin counts across all six site–layer combinations in the Abisko soil metagenomics dataset.

Species	**SF_Top**	**SF_Bot**	**KF_Top**	**KF_Bot**	**KJ_Top**	**KJ_Bot**
Bog-38 sp003162175	0	9	0	0	0	0
LD21 sp003141415	0	8	0	0	0	0
LD21 sp003141495	1	8	0	0	0	0
RBG-13-68-16 sp003133645	0	8	0	0	0	0
Fen-1308 sp003170655	4	0	0	0	0	0
Palsa-295 sp003153585	4	0	0	0	0	0
Palsa-465 sp003165655	4	0	0	0	0	0
Palsa-506 sp003168215	4	0	0	0	0	0
Palsa-295 sp003131985	3	0	0	0	0	0
Bog-38 sp003170935	0	3	0	0	0	0
FEN-979 sp003142515	0	3	0	0	0	0
Fen-1298 sp003142575	0	3	0	0	0	0
Fen-183 sp003152095	0	3	0	0	0	0
LD21 sp003141895	0	3	0	0	0	0
UBA12454 sp003142595	0	3	0	0	0	0
Fen-1467 sp903940425	0	0	0	0	0	2
Cryosericum hinesii	0	0	0	1	0	0
Deferrimicrobium sp937863105	0	0	0	1	0	0
Occallatibacter sp003133425	0	1	0	0	0	0
Palsa-1188 sp003132605	0	1	0	0	0	0

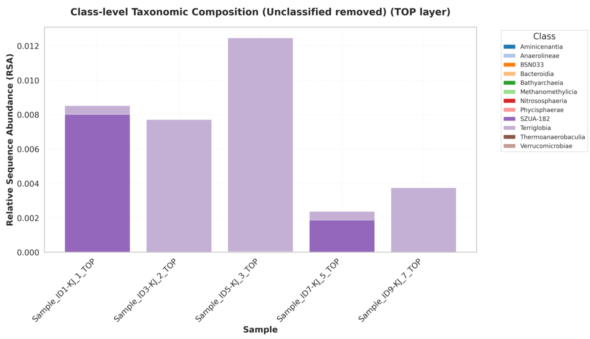
LD21 sp003141495 is the only species spanning more than one site–layer combination (1 in SF_Top, 8 in SF_Bot). The highest species counts occur in SF_Bot (*Bog-38 sp003162175*: 9; *LD21 sp003141415*: 8; *LD21 sp003141495*: 8; *RBG-13-68-16 sp003133645*: 8). SF_Top has four species at 4 bins each. KF_Bot contains two singleton species. KJ_Bot contains one species (2 bins). Both KF_Top and KJ_Top contain zero species. No species is shared across more than one location, indicating complete location-specific separation at the species level.



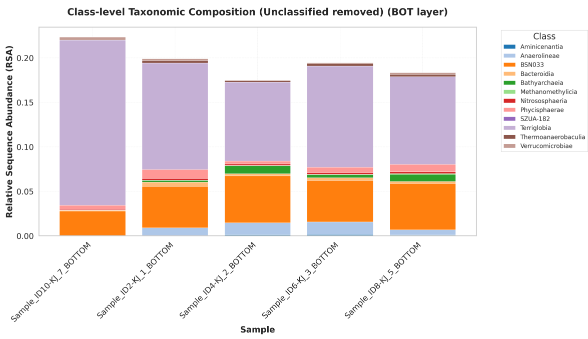
(a) SF TOP



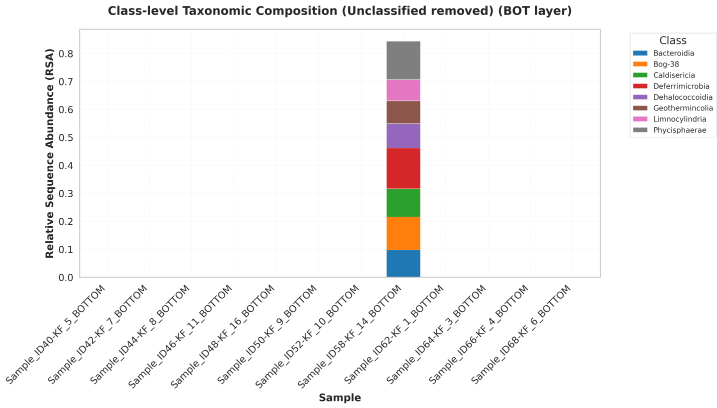
(b) SF BOTTOM



(c) KJ TOP

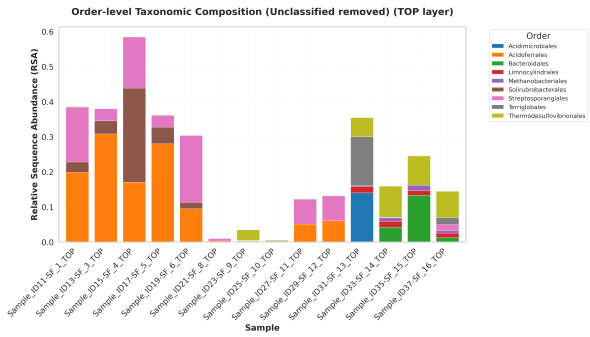


(d) KJ BOTTOM

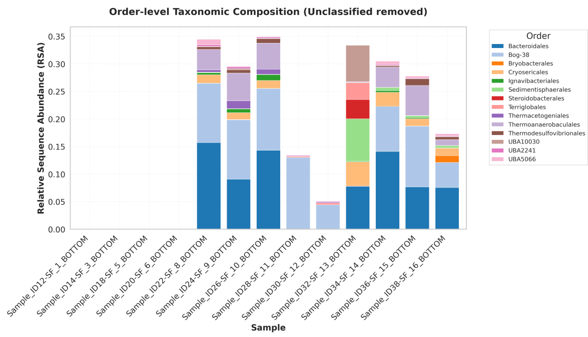


(e) KF BOTTOM

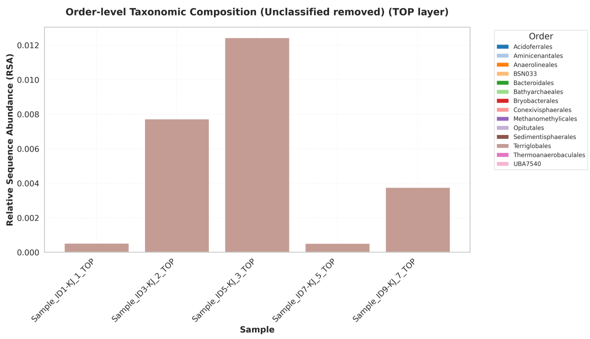
Figure 2: Class-level relative abundance distributions across all site-layer combinations showing the diversity of bacterial and archaeal classes in the Abisko soil samples.



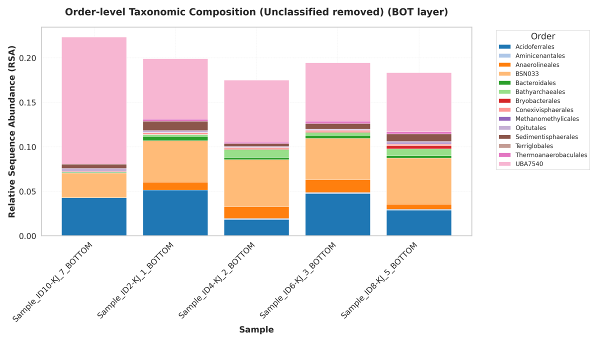
(a) SF TOP



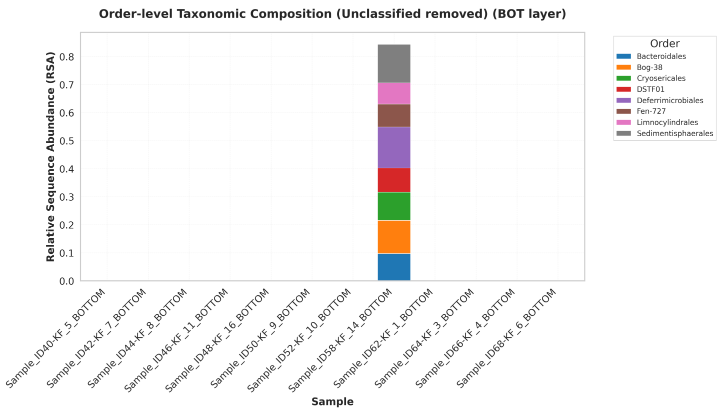
(b) SF BOTTOM



(c) KJ TOP

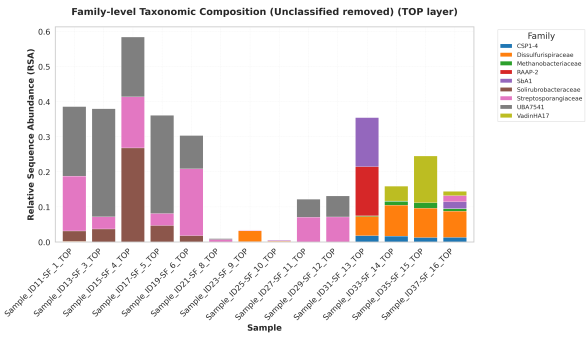


(d) KJ BOTTOM

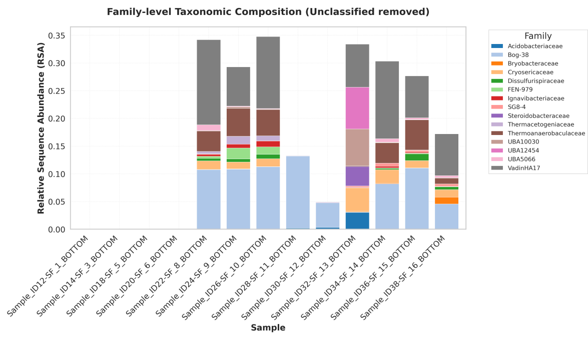


(e) KF BOTTOM

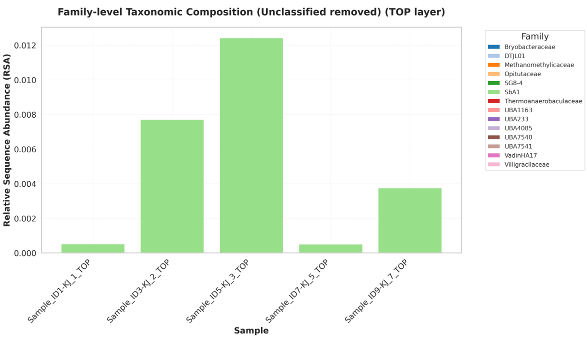
Figure 3: Order-level relative abundance distributions showing the taxonomic composition at the order rank across all sampling locations and depth layers.



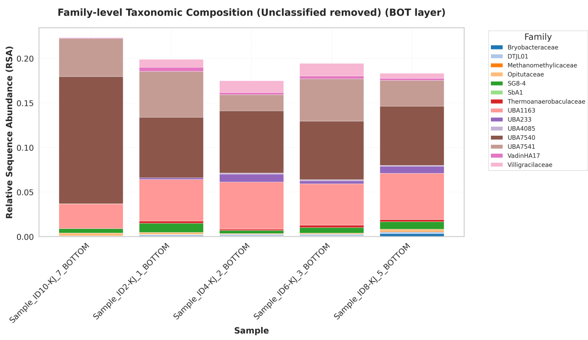
(a) SF TOP



(b) SF BOTTOM

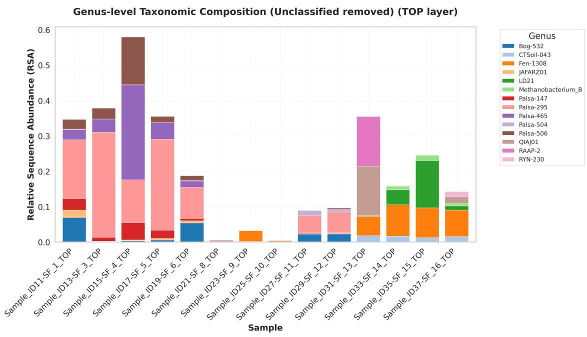


(c) KJ TOP

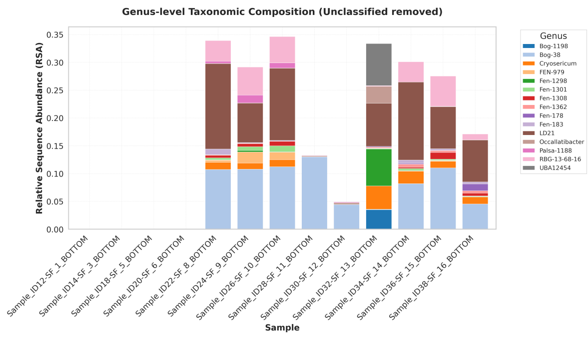


(d) KJ BOTTOM

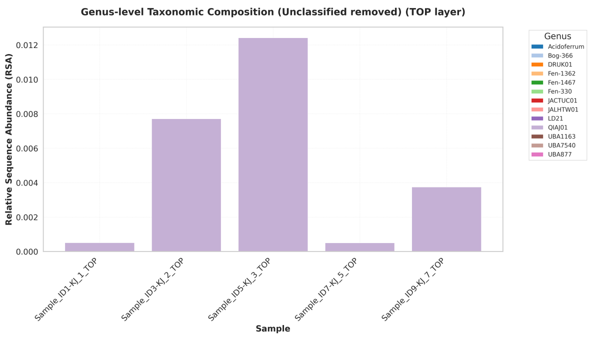
Figure 4: Family-level relative abundance distributions across sampling locations and depth layers. Each panel shows the percentage distribution of bins assigned to different families.



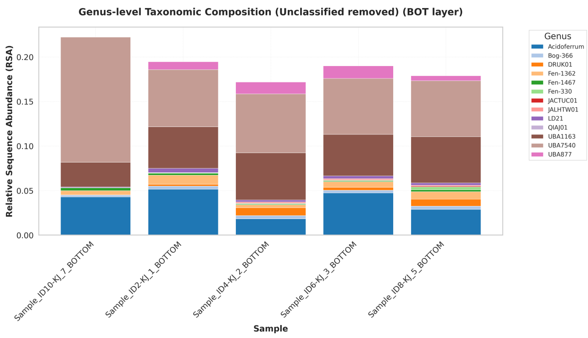
(a) SF TOP



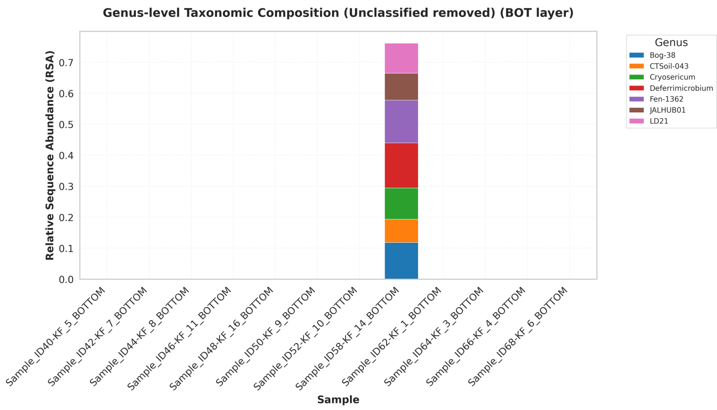
(b) SF BOTTOM



(c) KJ TOP



(d) KJ BOTTOM



(e) KF BOTTOM

Figure 5: Genus-level relative abundance distributions illustrating the fine-scale taxonomic structure and genera-specific patterns across different sampling sites and soil depth layers.